# SwitchSync Markets: A Preregistered Synthetic Stress Test of Switching Synchronization as a Financial-Network Mechanism

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**Abstract**

Dynamic link switching can induce stable synchronized states in sparse multilayer networks of chaotic oscillators [12]. We ask whether this mechanism admits a mathematically faithful and observationally identifiable translation to financial temporal networks, where cross-venue arbitrage channels open and close over time. Rather than fitting market data, we preregistered falsifiable gates and executed them *exactly once* on frozen synthetic systems: a two-layer FitzHugh–Nagumo (FHN) benchmark and a linear stochastic surrogate with paired switching schedules, under a frozen statistical decision contract (exact sign test plus a preregistered effect band). The calibrated demonstration gate passed: at $\sigma_{12} = 1.5$, $N = 40$, fast switching synchronizes the layers and slow switching does not. Every translation-relevant gate, however, failed to establish its claim. The switching-versus-static advantage was directionally consistent (8/8 positive paired differences, $p = 0.0078$) but below the preregistered magnitude band (INCONCLUSIVE/TIE); the strict comparison against the average graph and the best frozen static comparator is not interpretable under the frozen hierarchy, with diagnostically negative differences in all eight seeds. Because G1-weak did not pass, G2 was formally NOT_INTERPRETABLE; its diagnostic order statistic was non-significant. Robustness stages never exceeded their bands, and observational identifiability failed outright (precision and recall $\approx 0.35$ against a 0.6 bar), with asynchronous, last-observation-carried-forward sampling collapsing the contraction correlation from 0.948 to 0.010 (an Epps-like degradation [10]). We conclude that the financial translation is not supported for empirical or operational development under the preregistered criteria. This negative result does not refute the mathematics of the seed paper; it constrains what that mechanism can be expected to deliver as a measurable, decision-relevant object in financial data. No real market data were used.

## 1 Introduction

In a recent arXiv preprint, Eser, Medeiros, Riza and Engel showed that switching a sparse set of inter-layer links — rather than keeping them fixed — can stabilize inter-layer synchronization in two-layer networks of chaotic FitzHugh–Nagumo oscillators, with faster switching more effective [11, 12]. The mechanism is attractive to a financial modeler: markets trade the *same* economic asset on multiple venues; the cross-venue "links" (arbitrage channels, shared market makers, funding connections) are sparse, come and go over time, and venue prices are observed to lock together or drift apart episodically [1, 3, 17]. If switching-induced synchronization survived translation, "which

*Repository: `switchsync-markets`. This is a fully synthetic, preregistered study; no real market data were used at any stage. 

channels are open, in what order, and at what rate" would be a structural, measurable determinant of cross-venue price locking, distinct from aggregate connectedness measures [5, 7, 8].

That conditional is precisely what this paper tests, and the honest answer is negative under our preregistered criteria. Three failure modes make naive translations of synchronization results to finance unreliable: (i) fast-switching averaging theory implies that rapidly switched networks behave like their time-averaged counterpart [4, 18, 30], so an apparent "switching effect" may carry no information beyond the average graph; (ii) small ensembles invite effect-size inflation, so magnitude criteria must be fixed before observing data; and (iii) even a real mechanism is useless to empirical finance if it cannot be identified from asynchronously observed prices [10, 19, 29]. We therefore preregistered a gate hierarchy with frozen grids, seeds, estimands and decision rules; froze the code and configuration under content-addressed manifests; executed the suite exactly once under an auditable custody protocol; and committed the sealed outputs byte-for-byte.

Section 7 reports the outcomes (Table 1, Figure 1): the mechanism demonstrably operates in a calibrated synthetic setting (G0B PASS), but no translation-relevant gate established its claim, and identifiability failed. Throughout, we separate *confirmatory* preregistered outcomes, *diagnostic* quantities that the frozen hierarchy declares non-interpretable, *descriptive* observations, and *conjecture*.

## 2 Mathematical Background

**Synchronization and master stability.** For diffusively coupled identical oscillators, the stability of the synchronous state is governed by the transverse (conditional) Lyapunov exponents of the variational dynamics [26, 27]. On a static graph the problem factorizes into a master stability function (MSF) evaluated at the Laplacian eigenvalues.

**Time-varying coupling.** When the coupling network varies in time, the static MSF picture survives exactly only in special cases, e.g. commuting coupling matrices [6]. In the fast-switching regime, averaging results guarantee synchronization whenever the *time-averaged* network synchronizes [4, 30], with finite-rate deviations bounded by blinking-system theory [18]. Consensus theory gives the discrete-time counterpart: joint connectivity over time windows suffices [21, 24, 25, 28]. Empirically, rapidly switched links promote synchronization in a range of models [23, 32]; see [15] for a survey, [20] for the temporal-network formalism, and [22] for rigorous conditions under time-dependent diffusive coupling. *Crucially*, temporal networks *can* outperform any static network of equal resource budget in designed settings [31], so "switching never beats the average" is not a theorem; whether it holds in a given ensemble is an empirical question — the one our G1/G2 gates formalize.

**The seed mechanism.** The seed paper [12] couples two ring layers of chaotic FHN oscillators through a sparse, randomly re-drawn set of inter-layer links with dwell time $T_{\mathrm{swt}}$, and reports that sufficiently fast switching induces stable inter-layer synchronization at link densities where static sparse coupling fails; its predecessor [11] maps the edges of inter-layer synchronization under time-switched links.

## 3 Financial-Network Translation and Its Assumptions

The proposed reading maps: layers $\to$ venues quoting the same asset; inter-layer synchronization $\to$ cross-venue price locking (conceptually a cointegration-like common component [16]); the sparse switching link set $\to$ the currently active arbitrage/market-making channels; the switching rate $\to$

channel-rotation speed. Under this reading, price-discovery leadership that migrates between venues over time [1, 3, 17] is the observable trace of channel switching.

This translation embeds strong assumptions, each of which can independently fail without contradicting the seed mathematics:

1. **Mechanism transfer:** the contraction advantage of switching must survive in dynamics that are not two-layer chaotic FHN rings — here probed with a linear stochastic surrogate (Section 5).

2. **Beyond-average content:** the switching structure must matter *beyond* the time-averaged connectivity; otherwise standard aggregate connectedness measures [7–9] already capture everything, and correlation-based "regime" readings risk the classic confound between interdependence and heteroskedasticity [14].

3. **Identifiability:** the active channel set must be recoverable from observed prices, after removing common factors [2, 13] and despite asynchronous observation [10, 19, 29].

We emphasize what the translation, even if successful, would *not* give: a trading signal, a backtest, or any claim about market causality. None is made anywhere in this paper.

## 4 Hypotheses and Preregistered Gates

The preregistration (v8; hashes in Appendix A) freezes a gate hierarchy. Each gate has a frozen estimand, grid, seed block and decision rule; outcomes are PASS / FAIL / INCONCLUSIVE / EXECUTION_INVALID, with NOT_INTERPRETABLE for sub-gates whose interpretability is conditional.

**G0A (exact reproduction; expensive).** Reproduce the paper-scale regime ($\sigma_{12} = 0.1$, $N \in \{100, 200, 400\}$). *Not run* in this study (Section 7).

**G0B (calibrated demonstration).** At a re-calibrated coupling ($\sigma_{12} = 1.5$, $N = 40$), fast switching ($T_{\rm swt} \leq 10$) must synchronize and slow switching ($T_{\rm swt} \geq 160$) must not. *This is a qualitative demonstration of the mechanism, never an exact reproduction.*

**G0C (minimal MSF).** In a minimal two-node transverse-stability computation with switched anti-phase coupling channels, the stability onset must depend on $T_{\rm swt}$.

**G1-weak.** In the surrogate, the selected switching arm must beat the equal-instantaneous-density static-sparse comparator on paired evaluation seeds.

**G1-strict.** The arm must beat *both* the average-graph operator and the best static subset from a frozen candidate set ("best-of-frozen-candidate-set"; the admissible universe $\binom{24}{6} = 134{,}596$ is *not* searched, so no claim about the best admissible static is possible). Interpretable only if G1-weak passes.

**G2 (temporal order).** The ordered schedule must differ from the median over frozen order permutations of the same epoch multiset (permutation-median comparator + cross-seed sign test). Interpretable only if G1-weak passes.

**G3 (robustness).** The weak advantage must persist under implemented stresses (heterogeneity, directedness, signed weights); PASS requires the faithful and mild-heterogeneity stages to pass.

**G4 (identifiability).** A past-only estimator must recover the active channel set from noisy observations of the surrogate (precision and recall $> 0.6$), track contraction (correlation $> 0.5$, same-realization ground truth), and beat a factor-confounded baseline; tested under synchronous and asynchronous (LOCF) observation.

## 5 Synthetic Models and Experimental Design

**FHN benchmark (G0A/G0B).** Two ring layers of $N$ FitzHugh–Nagumo oscillators ($\epsilon = 0.05$, $a_{\mathrm{odd}} = 0.87$, $a_{\mathrm{even}} = 0.97$, rotational coupling angle $\phi = \pi/2 - 0.1$, intra-layer coupling $\sigma_{\mathrm{intra}} = 0.1$, nearest-neighbor ring) are coupled through a diffusive inter-layer term of strength $\sigma_{12}$ on a switching subset of $N_{\mathrm{IL}} = N/4$ mirror pairs, re-drawn uniformly at random every $T_{\mathrm{swt}}$ time units (RK4, $\mathrm{d}t = 0.01$). Synchronization is declared by a *deterministic* per-seed criterion: the tail time-average (final 25%) of the inter-layer error $E_{12}$ must fall below the frozen threshold 0.02 (this threshold is not an effect-band floor; see Section 6). G0B uses $N = 40$, $\sigma_{12} = 1.5$, total time 800, $T_{\mathrm{swt}} \in \{2, \ldots, 300\}$, five seeds per cell.

**Minimal MSF system (G0C).** A minimal $N = 2$ transverse-stability computation integrates the variational dynamics along the synchronized trajectory under smoothly switched anti-phase coupling channels ($\lambda_{\perp} = 2$, smoothing $\alpha = 5$), scanning $\sigma \in \{0.05, 0.15, 0.3, 0.5, 0.8\}$ and $T_{\mathrm{swt}} \in \{11, 25, 100, 135\}$. The anti-phase channel prerequisite is verified before any science value is computed.

**Linear stochastic surrogate (G1–G4).** To probe mechanism transfer outside chaotic FHN dynamics we use a linear stochastic surrogate: two coupled layers of $N = 24$ nodes whose inter-layer difference evolves through a sequence of linear difference-step maps (local spectral radius 1.04, i.e. intentionally near-unstable between couplings; coupling gain $\kappa = 0.6$ on open channels; weak intra-layer mixing 0.06). The estimand is the contraction rate $\gamma$ of the ordered product of the step maps over a frozen horizon $H = 240$: $\gamma > 0$ means the venue-difference (basis) process contracts. Schedules are *paired*: all comparators share the same base snapshots (six snapshots of six channel pairs) per seed, so differences isolate scheduling. The rate arms (fast / intermediate / slow) cycle the snapshot sequence 20/5/1 times within $H$. The switching arm and the best static candidate are selected on eight selection seeds and frozen *before* the eight disjoint evaluation seeds are opened; the static candidate set is the union of the base snapshots and a frozen 200-draw random search (248 candidates here).

**Identifiability testbed (G4).** On the same surrogate (horizon 600, fast dwell 6), a past-only, factor-free estimator based on venue differencing (window 8, observation noise 0.05) reconstructs the active channel set; it is compared against a deliberately factor-confounded level-correlation baseline (cf. [2, 13] for factor-removal rationale). Ground truth is the same realization's schedule. Observation is either synchronous (1:1) or asynchronous with last-observation-carried-forward at ratio 1:3 — an *asynchrony/LOCF (Epps-like degradation)* design, not a causal demonstration of the Epps effect.

**Custody.** All grids, seeds, tolerances and decision rules above were frozen in preregistration v8 and its bound execution contract v6 before execution; the code and configuration were sealed by a content-addressed freeze manifest (v8), and the suite ran exactly once under an authorization token whose SHA-256 is recorded in every artifact (Appendix A).

Table 1: Preregistered gate outcomes of the single authorized cheap-suite run (attempt `1baa47da06fede2a`, execution freeze v8). "Confirmatory" outcomes follow the frozen decision contract of Section 6; quantities marked "diagnostic" are reported but are, by the preregistered hierarchy, not gate outcomes. Zero technical failures, non-finite values or dropped seeds occurred in any gate.

| Gate | Question (frozen) | Verdict | Status |
|---|---|---|---|
| G0A | exact paper-scale reproduction | NOT_RUN | outside cheap-suite scope |
| G0B | calibrated small-$N$ demonstration | **PASS** | confirmatory (demonstration only) |
| G0C | minimal MSF $T_{\mathrm{swt}}$-dependence | INCONCLUSIVE | confirmatory |
| G1-weak | switching beats equal-density static | INCONCLUSIVE (TIE) | confirmatory |
| G1-strict | beats average graph and best frozen static | NOT_INTERPRETABLE | diagnostic only |
| G2 | temporal order beyond the multiset | NOT_INTERPRETABLE | diagnostic only |
| G3 | robustness of the weak advantage | INCONCLUSIVE | confirmatory |
| G4 | observational identifiability | **FAIL** | confirmatory |

# 6 Statistical Decision Contract

Every paired comparison uses one shared, frozen rule. For paired differences $d_1, \ldots, d_n$ ($n = 8$ evaluation seeds; differences with $|d| \leq 10^{-12}$ dropped and disclosed):

1. an exact two-sided sign test on the non-zero $d_s$ at $\alpha = 0.05$ (with 8/8 equal signs, $p = 2 \cdot \binom{8}{0}/2^8 = 0.0078$, so $n = 8$ *can* reach significance);
2. a preregistered *effect band* $b = \max\big(3\,\hat{\sigma}_{\mathrm{ddof}=1},\ \mathrm{floor}\big)$, with frozen floors: 0.02 for paired differences of the surrogate contraction rate $\gamma$, and 0.05 for the identifiability margin over the baseline (G4);
3. decision: if $p \geq \alpha \rightarrow$ INCONCLUSIVE (NOT_SIGNIFICANT); else if $\bar{d} > b \rightarrow$ PASS; else if $\bar{d} < -b \rightarrow$ FAIL; else $\rightarrow$ INCONCLUSIVE (TIE).

The G0B synchronization criterion (tail-averaged $E_{12} < 0.02$) is a *deterministic threshold* applied per seed, not an effect-band floor, and plays no role in the paired rule; the two 0.02 values are numerically equal but contractually distinct objects. A bootstrap interval (2,000 resamples, frozen seed) is reported as a *descriptor only* and plays no role in decisions. A global failed-seed policy invalidates any cell with more than 20% technical failures; the executed run had **zero** failures, non-finite values or dropped seeds. The hierarchy is frozen: G1-strict and G2 are interpretable only if G1-weak passes; otherwise their computed values are reported as diagnostics and are not gate outcomes.

# 7 Results

Table 1 and Figure 1 summarize the outcomes of the single authorized run (sealed attempt `1baa47da06fede2a`); Table 3 lists every decision statistic.

**G0B: PASS (confirmatory; calibrated demonstration only).** All fast cells ($T_{\mathrm{swt}} \leq 10$) synchronize (fraction 1.0) and all slow cells ($T_{\mathrm{swt}} \geq 160$) do not (fraction 0.2), with a non-increasing transition in between ($1.0 \rightarrow 1.0 \rightarrow 0.6 \rightarrow 0.2$ for $T_{\mathrm{swt}} = 20, 40, 80, 160$; Table 2, Figure 2). Five of five seeds succeeded in every cell. This demonstrates the switching-synchronization mechanism in a calibrated small-$N$ setting; it is *not* an exact reproduction of the paper regime and supports no claim about the $N$-dependence of the switching threshold.

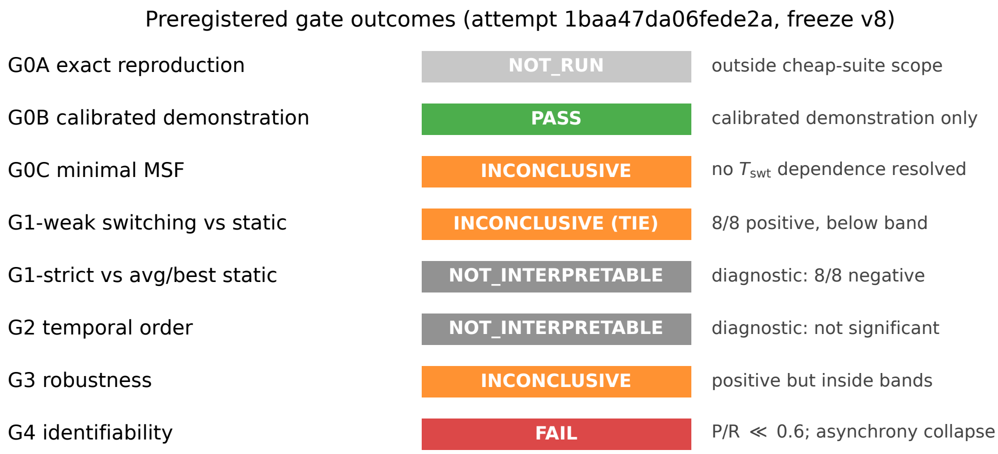


Figure 1: Gate summary. Grey chips mark quantities that the frozen hierarchy declares non-interpretable (diagnostic only).

Table 2: G0B calibrated demonstration ($\sigma_{12} = 1.5$, $N = 40$, $N_{\mathrm{IL}} = 10$, 5 seeds per cell, zero failures). The frozen criterion constrains only the fast band ($T_{\mathrm{swt}} \leq 10$: all fractions $\geq 0.5$) and the slow band ($T_{\mathrm{swt}} \geq 160$: all fractions $< 0.5$); intermediate cells are reported descriptively.

| $T_{\mathrm{swt}}$ | 2 | 5 | 10 | 20 | 40 | 80 | 160 | 300 |
|---|---|---|---|---|---|---|---|---|
| fraction synchronized | 1.0 | 1.0 | 1.0 | 1.0 | 1.0 | 0.6 | 0.2 | 0.2 |

**G0C: INCONCLUSIVE (confirmatory).** Transverse contraction exists: $\Psi < 0$ across the entire frozen grid (at the fastest $T_{\mathrm{swt}} = 11$: $\Psi = -0.008$ at $\sigma = 0.05$ down to $-0.177$ at $\sigma = 0.8$). But the stability onset is $\sigma = 0.05$ — the smallest grid value — at *every* $T_{\mathrm{swt}} \in \{11, 25, 100, 135\}$, so no switching-time dependence was resolved on the frozen grid (PARTIAL_NO_TSWT_DEPENDENCE). Descriptively, contraction is present; confirmatorily, the $T_{\mathrm{swt}}$-dependence that the translation needs was not shown at this resolution.

**G1-weak: INCONCLUSIVE / TIE (confirmatory).** The rate arm selected on the frozen selection seeds was *fast* (selection scores: fast +0.0366, intermediate +0.0071, slow −0.0155). On the eight disjoint evaluation seeds, all eight paired differences $\gamma_{\mathrm{arm}} - \gamma_{\mathrm{static}}$ are positive (sign test $p = 0.0078$), but the mean +0.0420 lies inside the preregistered band $b = 0.0666$ (Figure 3, left). Under the frozen rule this is INCONCLUSIVE (TIE): **directional evidence exists, but the effect failed the preregistered magnitude criterion.** We do not reinterpret this as a pass.

**G1-strict and G2: NOT_INTERPRETABLE (diagnostic only).** Because G1-weak did not pass, the frozen hierarchy renders G1-strict and G2 non-interpretable; we report their values strictly as diagnostics (Figure 3, middle and right; Table 3). Diagnostically, the G1-strict differences $\gamma_{\mathrm{arm}} - \max(\gamma_{\mathrm{avg}}, \gamma_{\mathrm{best\,static}})$ are negative in *all eight* evaluation seeds (mean −0.0046): the switching arm never beat the aggregate comparator in this ensemble. The G2 order statistic is small (+0.0030) and not significant ($p = 0.070$). These diagnostics must not be promoted to confirmatory conclusions — in either direction.

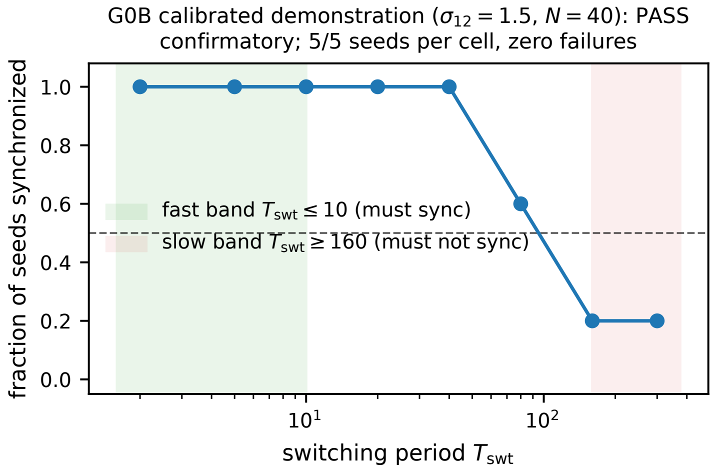


Figure 2: G0B synchronization fraction versus switching period (confirmatory). The frozen criterion constrains only the shaded bands.

**G3: INCONCLUSIVE (confirmatory).** Every stage shows a positive mean advantage $\gamma_{\text{fast}} - \gamma_{\text{static}}$ (from +0.0455 to +0.0631), with 8/8 positive signs in four of five stages — yet no stage exceeds its frozen band (Figure 4, Table 3), and the gate requires the faithful and mild-heterogeneity stages to pass. The signed stage satisfied its implementation prerequisites (per-seed negative weights present; Frobenius budget matched to $10^{-9}$), so the verdict is a clean INCONCLUSIVE, not an execution artifact.

**G4: FAIL (confirmatory).** The factor-free basis estimator recovers the active channel set with precision = recall = 0.353 under synchronous observation and 0.254 under asynchronous/LOCF observation — far below the preregistered 0.6 bar (Figure 5). It does beat the factor-confounded baseline (margin +0.162, band 0.065, 8/8 positive, $p = 0.0078$) and, under synchronous sampling, tracks contraction well (correlation 0.948). Under asynchronous/LOCF sampling the contraction correlation collapses to 0.010 — an Epps-like degradation [10, 29], consistent with the classical observation that asynchronous sampling destroys short-horizon comovement estimates even when consistent estimators exist [19]. Two of the four frozen conditions fail, so the gate fails.

**G0A: NOT_RUN (permanent decision).** The exact paper-scale reproduction ($\sigma_{12} = 0.1$, $N$ up to 400, 4,000 time units per cell) is outside the authorized cheap-suite scope and is recorded permanently as NOT_RUN — *outside the cheap-suite scope and unnecessary for the financial-translation decision.* It could be conducted only as a separate replication project; whatever its outcome, it cannot rescue G1–G4 (which concern translation, order effects and identifiability, not the mechanism's existence) and cannot authorize real-market research under the closed preregistration.

# 8 Why the Results Favor Aggregate Connectivity

The pattern of outcomes is exactly what fast-switching averaging theory predicts for this ensemble. The selected arm was the *fastest*; blinking and fast-switching results [4, 18, 30] say that in this regime the switched system inherits the stability of the time-averaged network, with finite-rate corrections.

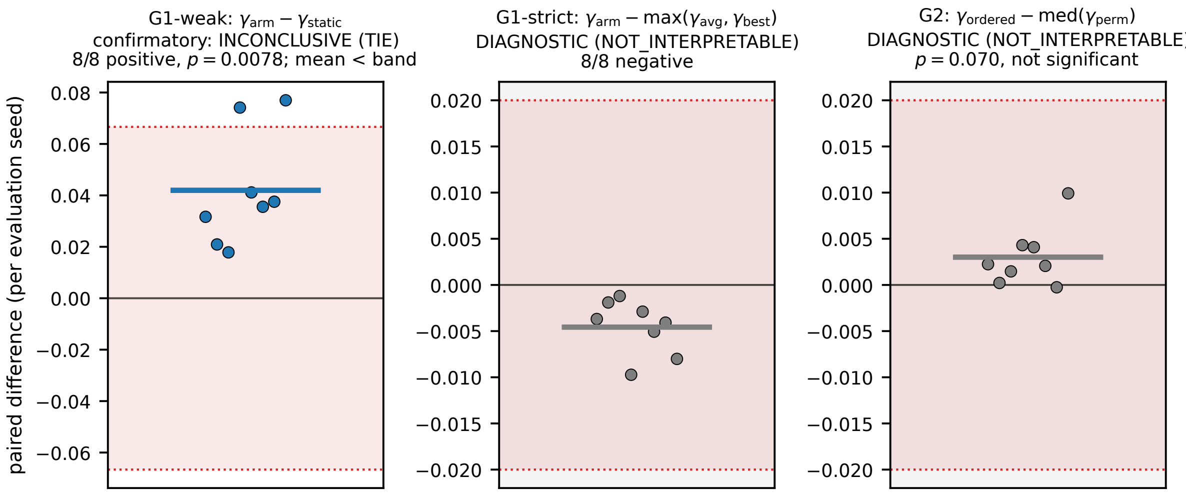


Figure 3: Per-seed paired differences with preregistered effect bands (red). Left: confirmatory G1-weak. Middle/right (grey background): diagnostic G1-strict and G2, NOT_INTERPRETABLE under the frozen hierarchy.

Consistently: the weak advantage over an equal-density *static sparse* graph is directionally positive (a static sparse subset is simply a worse average), while the diagnostic strict comparison against the average-graph operator is negative in every seed. Nothing in the confirmatory record shows switching delivering anything *beyond* its time-average here.

Two disciplined qualifications. First, this is not an impossibility claim: designed temporal networks can provably outperform static ones under resource constraints [31], and our frozen ensemble was not designed to sit in that regime; the negative reading applies to this preregistered ensemble, not to all switching systems. Second, the G2 permutation-median comparator aggregates signed order effects across seeds and can cancel sign-varying order effects; this limitation was frozen and disclosed in advance.

For finance the implication is direct: under this translation, measured "switching structure" carried no confirmed information beyond the average network. Aggregate connectedness measures [5, 7, 8] — and time-varying correlation models [9] with the usual heteroskedasticity discipline [14] — already target that average object. A switching-based market study would therefore need to demonstrate, first and preregistered, an effect beyond the average graph; ours did not.

# 9 Observational Identifiability and Asynchrony

Under synthetic conditions more favorable than those normally available in empirical finance — known dynamics, same-realization ground truth, a factor-free estimator by construction, and synchronous noise-controlled observations — channel-set recovery reached precision and recall of only 0.353. Beating the factor-confounded baseline (margin +0.162; the baseline's precision is 0.190) confirms that venue differencing removes the common factor, in the spirit of factor-removal approaches [2, 13], but "better than a confounded baseline" is far from "reliable". Under asynchronous/LOCF observation at a modest 1:3 ratio, recovery degrades further (0.254) and the contraction correlation collapses from 0.948 to 0.010. We stress the frozen terminology: this is an *asynchrony/LOCF (Epps-like degradation)* experiment — it reproduces the practical consequence

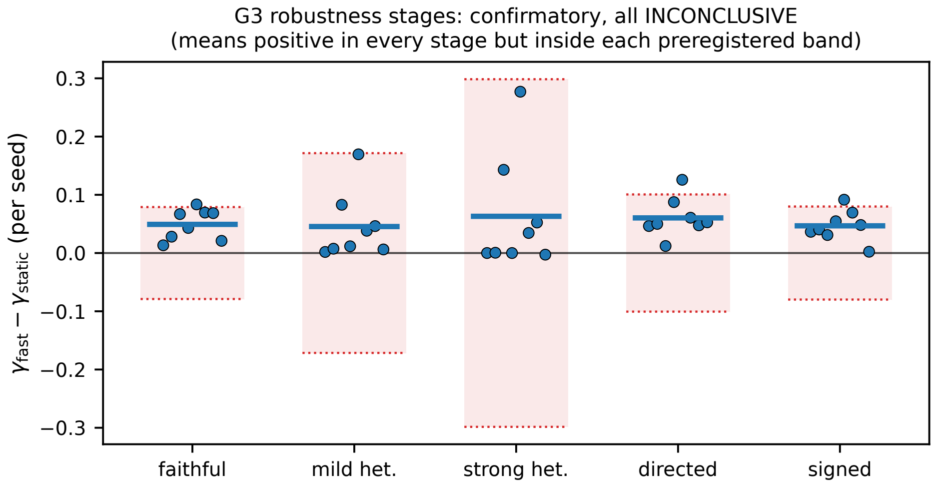


Figure 4: G3 per-stage paired differences with per-stage effect bands (confirmatory; all stages INCONCLUSIVE).

Table 3: Paired decision statistics ($n = 8$ evaluation seeds per test; exact two-sided sign test on non-zero differences; effect band $b = \max(3\,\hat{\sigma}_{\text{ddof}=1}, \text{floor})$; floors: 0.02 for surrogate $\gamma$, 0.05 for the identifiability margin). Rows marked † are diagnostic (NOT_INTERPRETABLE under the frozen hierarchy).

| Test | mean $\bar{d}$ | band $b$ | signs (+/−) | $p$ | verdict |
|---|---|---|---|---|---|
| G1-weak: $\gamma_{\text{arm}} - \gamma_{\text{static}}$ | +0.0420 | 0.0666 | 8/0 | 0.0078 | INCONCLUSIVE (TIE) |
| G1-strict†: $\gamma_{\text{arm}} - \max(\gamma_{\text{avg}}, \gamma_{\text{best}})$ | −0.0046 | 0.0200 | 0/8 | 0.0078 | (diagnostic) |
| G2†: $\gamma_{\text{ord}} - \text{med}(\gamma_{\text{perm}})$ | +0.0030 | 0.0200 | 7/1 | 0.0703 | (diagnostic) |
| G3 faithful | +0.0492 | 0.0789 | 8/0 | 0.0078 | INCONCLUSIVE (TIE) |
| G3 mild heterogeneity | +0.0455 | 0.1715 | 8/0 | 0.0078 | INCONCLUSIVE (TIE) |
| G3 strong heterogeneity | +0.0631 | 0.2984 | 6/2 | 0.2891 | INCONCLUSIVE (NOT_SIG.) |
| G3 directed | +0.0603 | 0.1006 | 8/0 | 0.0078 | INCONCLUSIVE (TIE) |
| G3 signed | +0.0467 | 0.0799 | 8/0 | 0.0078 | INCONCLUSIVE (TIE) |
| G4 beats-baseline margin | +0.1625 | 0.0646 | 8/0 | 0.0078 | PASS (condition; gate FAILs on P/R) |

highlighted by Epps [10] in our setting; it is not a causal demonstration of the Epps effect, and estimators designed for non-synchronous data [19, 29] were not the object under test. Since real markets add unknown dynamics, latent factors, microstructure noise and far harsher asynchrony, the preregistered conclusion is that the mechanism's observational signature is not reliably identifiable in this class of designs; a real-data "detection" built on such an estimator would not be interpretable.

# 10 Limitations

1. **Scope of the negative result.** All conclusions are relative to the frozen ensemble, grids and decision contract. We did not search parameter space for regimes where switching beats the average [31]; that would be a different, new preregistration.

2. **Small $n$, wide bands.** With $n = 8$ paired seeds and a $3\hat{\sigma}$ band, magnitude criteria are conservative; consistent directional effects (8/8 signs) can land INCONCLUSIVE, as G1-weak

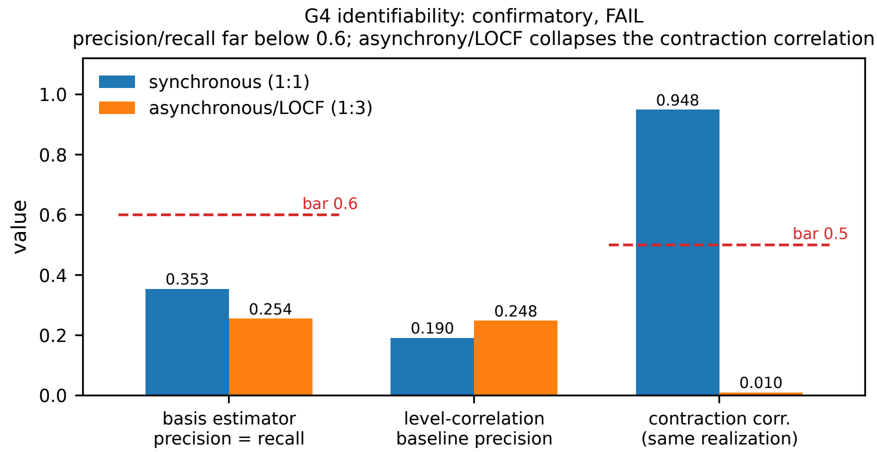


Figure 5: G4 identifiability metrics under synchronous (1:1) and asynchronous/LOCF (1:3) observation (confirmatory; gate FAIL).

and four G3 stages did. This is by design (anti-inflation), but it limits power.

3. **Comparator strength.** G1-strict uses the best of a frozen 248-candidate static set, not the best admissible static subset ($\binom{24}{6} = 134{,}596$); claims about the true static optimum are impossible and none are made.

4. **G2 statistic.** The permutation-median comparator detects a systematic displacement from the permutation median; sign-varying order effects across seeds can cancel (frozen, disclosed limitation).

5. **G0C resolution.** The onset grid starts at $\sigma = 0.05$ and the onset saturated at that smallest value for every $T_{\mathrm{swt}}$; a finer grid might resolve dependence. The minimal $N = 2$ construction is also the weakest link to the full spatiotemporal system.

6. **G0A not run.** The exact paper-scale reproduction was not executed; G0B is a calibrated demonstration at different coupling and size and cannot substitute for it.

7. **The surrogate is not a market model.** It is a mechanism testbed. No claim about real markets, trading, price prediction or market causality follows from any result here — positive or negative. No real market data were used anywhere.

## 11 Reproducibility and Audit Trail

The study enforced preregistration-grade custody end to end: (i) scientific preregistration and operational execution contract, versioned and bound by canonical hashes (v8/v6 at execution); (ii) a content-addressed freeze manifest over all code, configuration, tests and documents (117 files), whose stored content hash equals independent recomputation; (iii) a single authorized execution from the freeze commit with a clean tree, identified by an authorization token recorded only as its SHA-256, producing attempt `1baa47da06fede2a`; (iv) a sealed attempt directory whose manifest inventories every artifact by size and SHA-256 and whose seal equals the manifest content hash; (v) a separate post-execution artifact audit that reconstructed the package and recomputed all identities, hashes and decision statistics exactly; (vi) byte-for-byte custody of the sealed outputs in the repository under an annotated tag, distinct from the execution freeze commit. Every figure and table in this paper is generated from the frozen JSON reports without re-running any simulation. Earlier execution freezes v2–v7 were each found non-executable by separate pre-execution artifact audits and were *never run*; consequently no result predates the sealed attempt. All identifiers appear in Appendix A.

## 12 Conclusion

SwitchSync Markets demonstrates the switching-synchronization mechanism in a calibrated synthetic setting (G0B PASS), but under the preregistered criteria it does not establish an advantage beyond aggregate connectivity (G1-weak INCONCLUSIVE/TIE; G1-strict diagnostic and non-interpretable), a robust temporal-order effect (G2 non-interpretable; G3 INCONCLUSIVE), or reliable observational identifiability (G4 FAIL). The financial translation is therefore not supported for empirical or operational development under the preregistered criteria: no predictive or operational market study is justified, and no trading signal, backtest, PnL, Sharpe or deployment claim exists or is implied. No real market data were used. This outcome does not refute the mathematical results of the seed paper [12]; it delimits what that mechanism can currently be expected to deliver as a measurable, decision-relevant object in financial data, and it documents — with full custody — how a preregistered negative result of this kind can be produced and audited.

## Transparency Statement

Generative-AI systems assisted with code generation, testing, artifact auditing, and manuscript drafting. The human author reviewed the scientific claims and assumes full responsibility for the submitted work.

## A Audit and Reproducibility Appendix

**Identifiers.** Table 4 lists the immutable hashes. The execution scope was `cheap-suite`; G0A is permanently NOT_RUN. The authorization token is recorded only as its SHA-256:

`3b4b903518fa5cb337cbf5416aa8cc6db8db47c1548b39545f562158b07aca29`

the attempt id satisfies $\text{attempt} = \text{sha256}(\text{campaign} \,\|\, \text{scope} \,\|\, \text{token\_sha})[:16]$, so an auditor recomputes it without the raw token. Zero failure records, non-finite values or dropped seeds occurred. No real data were accessed.

**Seed blocks (frozen in prereg v8).** chaos $\{7\}$; G0A $\{11, 12, 13\}$; G0B $\{11, \ldots, 15\}$; G0C $\{31\}$; paired selection $\{41, \ldots, 48\}$; paired evaluation $\{81, \ldots, 88\}$; G3 stages $\{51, \ldots, 58\}$; identifiability $\{61, \ldots, 68\}$.

**Environment (pinned and verified at execution).** Python 3.14.3; numpy 2.4.4; scipy 1.17.1; matplotlib 3.10.9; Linux x86-64 (glibc 2.35); RK4 with $\mathrm{d}t = 0.01$; runtime $\approx 8\,\mathrm{min}\,44\,\mathrm{s}$ for the full cheap suite.

Table 4: Immutable identifiers of the preregistration, the execution freeze and the sealed result. Earlier execution freezes v2–v7 were each found non-executable by separate pre-execution artifact audits and were *never run*; no result predates the attempt below.

| Object | SHA-256 / identifier |
|---|---|
| prereg v8 (canonical) | `a667434d0d7c5905d62cf81b79699f2799e2597128631cab1e4792bed6053dd9` |
| prereg v8 (file) | `896163891b8b43f42511a1a57294d6075f3b65f33877780daf82cc03a54d568e` |
| execution contract v6 (canonical) | `6dbe79ddf1d0b6e5e63f74a5ae3ed947abfc837419f4b7938fdcf8c09f525ee8` |
| execution contract v6 (file) | `80609ef5a90817813f0a9978ec017ab00734c3681abb0429fbe4ec4582b8c2bb` |
| execution freeze v8 (content hash) | `2991e064ced6154c149fa57d4fc77e855bede51750565c264972b369778d112d` |
| freeze commit (= runtime HEAD) | `5ca40c7da87f583c7cea1cefd00cc978c3731d98` |
| result-custody commit | `48339e47d3a0911949cf25c83a50e63fc9572aa5` |
| campaign / attempt id | `44be3a5d07e67b5e` / `1baa47da06fede2a` |
| manifest content hash (= SEALED) | `4e029d814437695d99c4b5cc62cbe3cf94ebd0e46c6e9ae51d54cc5f28f3c33d` |
| `g0b_calibrated_v2.json` | `d592b75d59b6e50c51ec6263c0a8469654651e0ef247fd2fadd3cf5adad4d4e5` |
| `g0c_msf_v2.json` | `150a53dc42236742d347012704a00a4d2e1db52d99bb800341bbc3ec6de484d0` |
| `g1_g2_paired_v2.json` | `b6242cd72510ce6d8b0daa5edba31b1c124f3fa519eae92ac0a408f06decbce0` |
| `g3_robustness_v2.json` | `2f16a4578c5b48cbb8621455b5a37042ec615a7127c9d709acb43ab1a3414964` |
| `g4_identifiability_v2.json` | `bd148d1ed23ba057df41dedfbdb7414b7d450a7f5c28b59617e354d5bbb597e6` |

**Execution history declaration.** Execution freezes v2–v7 were each found non-executable by separate pre-execution artifact audits (defects ranged from custody flaws to a missing frozen tolerance block) and were superseded without ever producing a result. The sealed attempt `1baa47da06fede2a` under freeze v8 is the *only* scientific execution of this project.

# B Frozen Decision Details

**Shared rule.** Exact two-sided sign test on non-zero paired differences at $\alpha = 0.05$; effect band $b = \max(3\hat{\sigma}_{\mathrm{ddof}=1}, \mathrm{floor})$; PASS/FAIL only outside $\pm b$; bootstrap (2,000 resamples, frozen seed 20260713) reported as descriptor only.

**G1/G2 selection protocol.** Arm and best-static comparator selected on the selection seed block only, using a common per-seed validity mask over *all* rate arms and *all* 248 static candidates, then frozen before evaluation seeds are opened. Ties break canonically (arm order fast<intermediate<slow; lexicographic subsets).

**G4 conditions (all four required).** mean precision $> 0.6$; mean recall $> 0.6$; same-realization contraction correlation $> 0.5$; significant positive margin over the factor-confounded baseline (floor 0.05). Observed: 0.353; 0.353; 0.948 (synchronous); margin +0.162 (PASS) — two conditions fail, hence gate FAIL.

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